

Phase 0

“A library is not a luxury but one of the necessities of life.” – Henry Ward Beecher

3. Introduction

In today's information era, public libraries remain a pillar of knowledge, learning, and socialization. However, with the need for faster, more organized and easily accessed services increasing, the need for an efficient management system has become obvious.

Public Library Management System

Is designed to overcome such problems by automating core processes such as cataloguing, member registration, book circulation and fine management.

This system will replace traditional manual processes with a computer-based system with a view to enhancing precision, time efficiency and friendliness. It can aid librarians in resources management with ease while offering the users seamless access to information and services.

The project scope includes the provision of physical and digital resources, providing convenient access search capabilities, tracking borrowing history, and generating reports to enhance decision-making. Through these additions, the system enables the library to meet modern demands without compromising its cultural and educational personal.

Lastly this project is more about the library it's not so much about technology, it's about preserving the library's role as a gateway to knowledge while adapting it to the need of digital age.

4. PROJECT OVERVIEW

Nowadays we are living in technology-driven world whereby libraries remain an important part of communities while giving access to knowledge, learning, and opportunity. Although many public libraries still depend on manual systems that manage books, and staff are time-consuming and inefficient. The project will design and develop a system that will perform/bring simplicity, accuracy, and efficiency. Another main reason for the project is to

develop a system that will assist/allow public librarians to manage book information, update existing records, add new books, and track book availability. The librarians will benefit from a project as it will be improving their experience and at the same time it will be limiting errors and automating services. The project aim is not on books, users, or certain library staff only, but it will also help staff management to process salary information, arrange appointments, and record staff leave. The aim of the project is to improve the operations of public libraries by erasing manual processes with the intention of activating technology-driven processes. This public library management system project will support developing communities educationally as it intends to give the required educational-related resources.

5. Problem Statement

1.Poor Data Management

Most library management systems suffer with accuracy of data resulting in such problems like duplication, inconsistencies and old data. This might lead to confusion when searching for a book check availability or borrowing history. Our system will help by automating the update and connecting with outside databases for more accurate information.

2.Failure to Update in Real Time

There are also systems that do not keep information up to date. A perfect example would be where a book is labelled "available" but it might not be available because the system is not updated. Library Management systems help by tracking and synchronizing changes so that availability is always up to date

3. Security Issues

Library systems contain private user data, such as personal information and borrowing history. Without proper security mechanisms such the information can be susceptible to exploitation by unauthorized users. The system can ensure security through the use of user authentication, data encryption and automated backup to protect information.

6. SDLC Phases

1. **Planning**

Uni Track System aims to manage book information by enabling book search by title, author, genre or ISBN. It also focuses on managing library staff information such as tracking staff leave, scheduling appointments and processing of salaries. The system will also automate staff leave and users will be able to borrow and return books without the assistance of the librarian.

2. **Analysis**

Interviews conducted with the library staff and users to ensure system requirements and identifying areas that need improvement. Functional requirements for book management include adding, searching, updating and deleting book details. It also involves issuing and returning of books and managing fines, the system also includes registering and managing library members/patrons. Functional requirements for library staff management involve processing of staff leave requests, managing appointment schedules and salary records.

Non-functional requirements for book management include ensuring that data is protected, creating user-friendly websites to give users easier access to the library system and response time to user actions within 2 seconds for basic operations. Non-functional requirements for staff management include role-based access control.

3. **Design**

The Uni Track System will design the architecture that corresponds with the documented requirements from the analysis phase, and this system design involves user interfaces for book information and managing staff (leave, appointments, salary). Database design ensures integrity and accessibility by allocating tables for books, staff and transaction records. Security design including data protection and ensuring suitable role-based access controls are part of the design.

4. **Development**

In the development stage, we will need to write an actual code that will implement attributes for managing library staff information (including leave, appointments and payroll) and book information. This includes building a book management module to handle all properties regarding adding, searching, updating and deleting book records and developing a staff

management module that supports leave requests, appointment scheduling and salary record management.

7. Engagement with the community as part of the topic investigation.

Public Library Management System

Public libraries management system in South Africa offers as essential community Centre, offering passage to information, learning, and cultural improvement., nevertheless librarians over the nation face multiple challenges that obstruct their capability to deliver productive assistance. The following report synthesizes insights from previous studies and discussions with librarians to emphasize the major challenges and introduce promising resolutions.

Major Obstacles

1. Insufficient Materials and Infrastructure

Numerous public libraries function with outdated resources and insufficient utilities. In Kwa-Zulu Natal Province, for example, libraries compete with cabins, pour databases, and inadequate internet connectivity.

2. Digital Divide

Entry to technology persists a crucial block, particularly in non-urban areas. The digital inequality worsens disparities, restricting users' capability to access internet resources and information.

3. High Initial Costs

Implementing a computerized structure demands important investment in hardware components, software licenses, and networking infrastructure. This can be a challenge, particularly for libraries with insufficient budgets.

8. User Requirements

Guest

- Login into the system securely
- Guest must be able to borrow and return books
- Place reservations on borrowed items

- Receive notification when held items are ready for pick up
- Guests can search for books by title, author or subject
- Pay fees or fines via online payment

Admin

- Generate reports on borrowed books and overdue books
- Display **overdue** fines and damages fees
- Track due dates of the borrowed books
- Send notifications that's reminds for overdue or approaching returns
- Give tasks or shifts to staff
- Remove old books and add new ones
- Manage library staff
- Manage book information like title, author or location

Non-Functional requirements

- Local pages quickly and response time
- The system must be able to handle many users at the same time
- Personal data should be protected from hackers
- The system must be easy and simple to use on both guest and admin

9. Functional Requirements

1. User Registration and Authentication

2. User Roles

3. Book Management

4. Search and Browse

5. Borrowing and Returning

6. Reservation

7. Overdue and Fine Management

8. Reporting

9. Notification System

10. Digital Resource Access (Optional)

Non-Functional Requirements

1. Performance

2. Scalability

3. Availability

4. Security

5. Usability

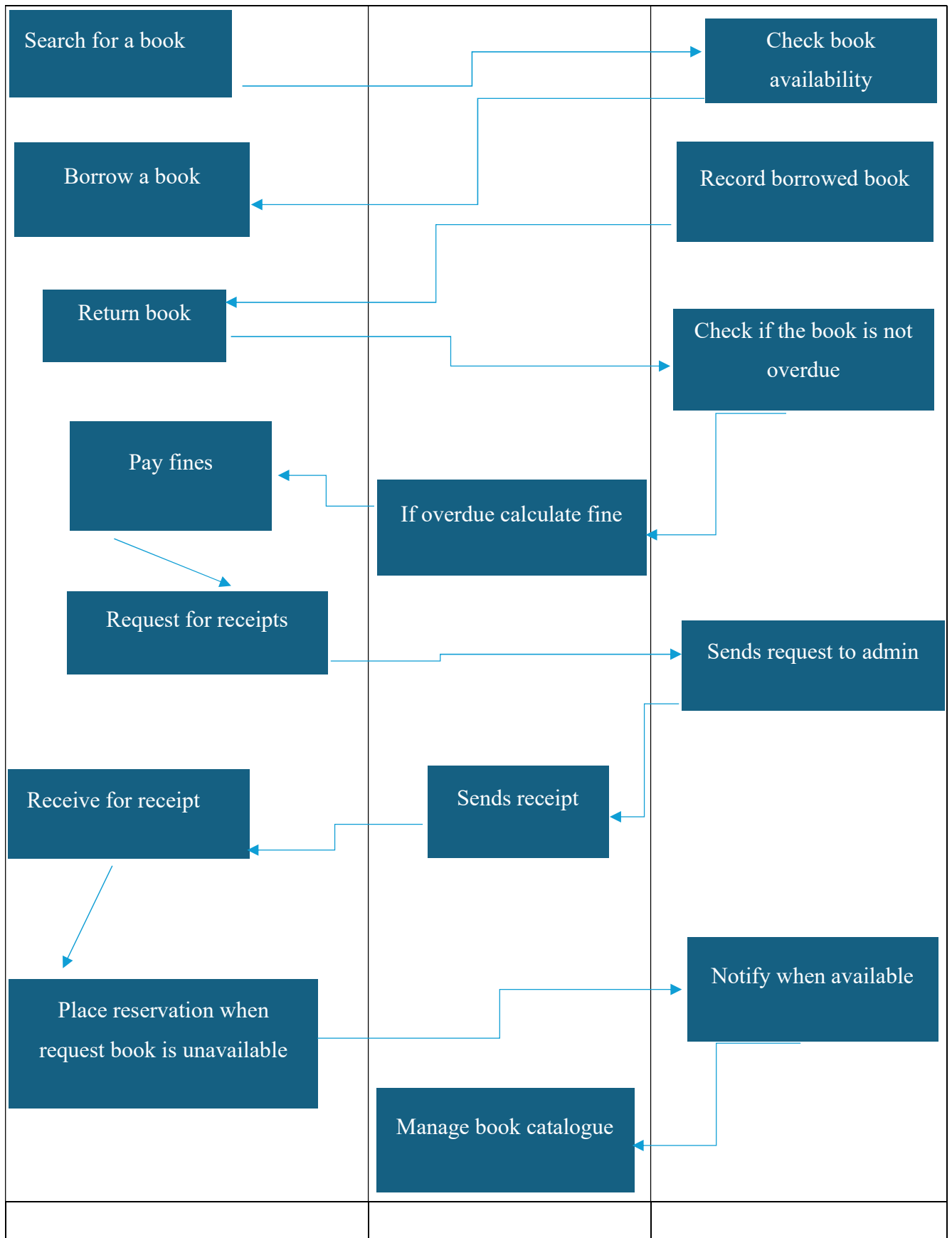
6. Maintainability

7. Portability

8. Backup and Recovery

11. Activity Diagram

Guest	Admin	system
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13. Use Case Description table

User case name:	Registering a member	
Scenario:	A librarian wants to check if a member is registered	
Triggering event:	A person requests to be a member of the library	
Brief description:	This use case table will show the process of adding a new member to the system so that they can easily access the library services once they are a member.	
Actors:	Librarian, New Member	
Related Use Cases:	Store data member data securely	
Stakeholders:	Library	
Preconditions:	1. A user must provide their personal details 2. The member must get a card that official makes them a member of the club 3. The member's status is set to active to show that they are active as a member.	
Post Conditions:	1. A member's account is created in the library system 2. The member receives a library card	
Flow of activities:	Actor	System
	1.A new member request membership in the library 2.Librarian takes personal details of the new member 3.The librarian enters the details into the system.	1. The system will capture the personal details of the new member. 2. The system verifies the information

	4. Librarian issues out the membership card to the new member	3. The system set a status member to “active”. 4. System will generate a new membership card of the new member.
Expectations	Members must have their own library card Protect a member’s personal information from unauthorized users.	

14. Prototypes

